MTE 204 Previous Midterm Test Questions

Question

Solve the system of linear equations below using Gauss elimination with partial pivoting.

$$\begin{aligned} x_1 + x_2 - x_3 &= -3 \\ 6x_1 + 2x_2 + 2x_3 &= 2 \\ -3x_1 + 4x_2 + x_3 &= 1 \end{aligned}$$

Solution

Express the system of equations in matrix form

$$\begin{bmatrix} 1 & 1 & -1 \\ 6 & 2 & 2 \\ -3 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

Place in the form of an augmented matrix for more straightforward elimination

$$\begin{bmatrix} 1 & 1 & -1 & -3 \\ 6 & 2 & 2 & 2 \\ -3 & 4 & 1 & 1 \end{bmatrix}$$

Begin forward elimination of the first column. First, we pivot by swapping rows 1 and 2

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 1 & 1 & -1 & -3 \\ -3 & 4 & 1 & 1 \end{bmatrix}$$

Elimination formula for row 2 is

$$\text{row } 2 = \text{row } 2 - \frac{a_{21}}{a_{11}} \text{row } 1$$

$$\text{row } 2 = \text{row } 2 - \frac{1}{6} \text{row } 1$$

Elimination formula for row 3

$$\text{row } 3 = \text{row } 3 - \frac{a_{31}}{a_{11}} \text{row } 1$$

$$\text{row } 3 = \text{row } 3 - \frac{-3}{6} \text{row } 1 = \text{row } 3 + \frac{1}{2} \text{ row } 1$$

Applying these elimination equations yields the following

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 0 & {}^{2}/_{3} & -{}^{4}/_{3} & -{}^{10}/_{3} \\ 0 & 5 & 2 & 2 \end{bmatrix}$$

Continue elimination to the second column. First, we pivot by swapping rows 2 and 3

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 0 & 5 & 2 & 2 \\ 0 & {}^{2}/_{3} & -{}^{4}/_{3} & -{}^{10}/_{3} \end{bmatrix}$$

The elimination formula for row 3 is as follows

$$\text{row } 3 = \text{row } 3 - \frac{a_{32}}{a_{22}} \text{row } 2$$

$$\text{row } 3 = \text{row } 3 - \frac{{}^{2}/_{3}}{5} \text{row } 2 \ = \ \text{row } 3 \ - \frac{2}{15} \text{row } 2$$

Applying this elimination formula yields

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 0 & 5 & 2 & 2 \\ 0 & 0 & -{}^{8}/_{5} & -{}^{18}/_{5} \end{bmatrix}$$

Placed in the standard matrix form

$$\begin{bmatrix} 6 & 2 & 2 \\ 0 & 5 & 2 \\ 0 & 0 & -{}^{8}/_{5} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -{}^{18}/_{5} \end{bmatrix}$$

Applying back substitution yields the following

$$x_3 = \frac{-\frac{18}{5}}{-\frac{8}{5}} = 2.25$$

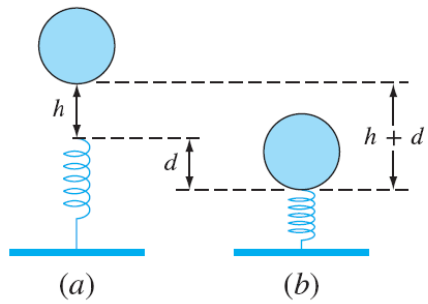
$$x_2 = \frac{1}{5}(2 - 2x_3) = \frac{1}{5}\big(2 - 2(2.25)\big) = -0.50$$

$$x_1 = \frac{1}{6}(2 - 2x_3 - 2x_2) = \frac{1}{6}\big(2 - 2(2.25) - 2(0.5)\big) = -0.25$$

Question

A mass m is released a distance h above a non-linear spring. The resistance force of the spring F is defined with respect to the compression of the spring d as shown below

$$F = -(k_1 d + k_2 d^{3/2})$$



The conservation of energy between the undropped mass (state a) and the fully compressed spring (state b) is given by:

$$E_{potential} = E_{spring}$$

$$mgd + mgh = \frac{2k_2 d^{5/2}}{5} + \frac{1}{2}k_1 d^2$$

For this problem, use the following parameters: k_1 = 40,000 g/s^2, k_2= 40 g/(s^2 m$^{0.5}$), m = 95 g, g = 9.81 m/s^2, and h = 0.33 m.

a) Estimate the value of *d* using the graphical method. Include the plot used to estimate d in your solution. (hint: substitute x values of 0.0 and 1.0 as a start to generate a plot)

b) Solve for *d* using the secant method, with an initial guess of *d*=1.0 m and a second initial guess from your graphical solution in part a. Show calculations for the first 3 iterations and calculate the relative percent error ε_a.

Solution

Part A

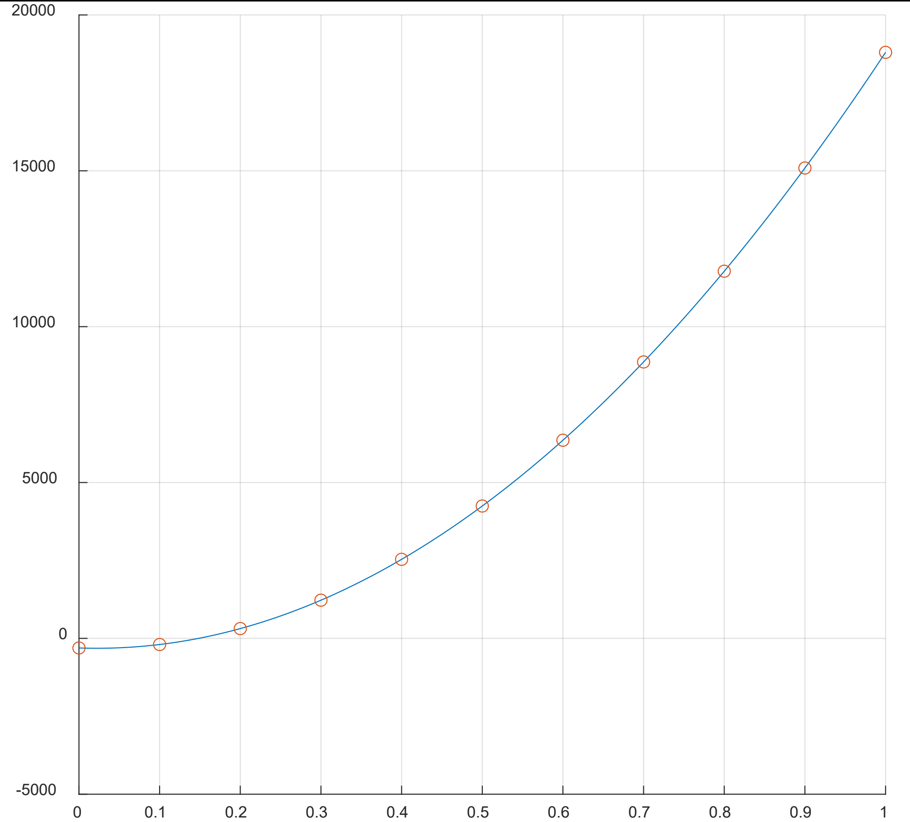
We begin by placing reorganizing the conservation of energy equation so that the equation equals zero

$$0 = \frac{2k_2 d^{5/2}}{5} + \frac{1}{2}k_1 d^2 - mgd - mgh$$

$$0 = 16d^{\frac{5}{2}} + 20000d^2 - 931.95d - 307.5435$$

Perhaps the easiest way to sketch this function by hand is to create a table of several values between d=0 and d=1. This will allow one to estimate where the root occurs. An example of this is provided below with the actual plotted function. From these evaluated points, we know our root is between 0.1 and 0.2. Let us estimate that the root occurs at $d = 0.15$. The actual root value is 0.148292.

d	$f(d)$
0	-307.5
0.1	-196.7
0.2	314.0
0.3	1225
0.4	2536
0.5	4247
0.6	6357
0.7	8868
0.8	11 780
0.9	15 090
1.0	18 800



Part B

The update equation for the secant method is as follows

$$x_{i+1} = x_i + \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

The relative percent error used for the secant method is

$$\epsilon_a = \frac{x_{i+1} - x_i}{x_{i+1}} * 100\%$$

We will initialize the secant method with $d_{-1} = 1$ and $d_0 = 0.15$. The first three iterations of the secant method for these initial values are listed in the table below.

Iteration	x_{i-1}	x_i	$f(x_{i-1})$	$f(x_i)$	x_{i+1}	$f(x_{i+1})$	ϵ_a
1	1	0.15	18 880	8.6640	0.1496	6.6652	0.2619%
2	0.15	0.1496	8.6640	6.6653	0.1483	0.0444	0.8812%
3	0.1496	0.1483	6.6653	0.0444	0.14829	0.0002	0.00591%

The actual root is 0.1482924. Solutions from students will be evaluated against the d_0 they have determined from their graphical method.

Question

For the following three sets of linear equations:

a) Identify which set(s) would not converge using the naive Gauss-Seidel algorithm based on the convergence criterion demonstrated in class. Do not change the ordering of the equations for this part.

b) For the equation set(s) that do not meet the convergence criterion, apply pivoting and demonstrate if this leads to convergence.

Set 1	Set 2	Set 3
$9x + 3y + z = 3$	$x + y + 6z = 8$	$-3x + 4y + 5z = 6$
$-6x + 8z = 8$	$x + 5y - z = 5$	$-2x + 2y - 4z = -3$
$2x + 5y - z = 6$	$4x + 2y - 2z = 4$	$2y - z = 1$

Solution

Part A

There are two criteria for convergence using the naive Gauss-Seidel algorithm. 1) Diagonal elements must be non-zero, and 2) the matrix must be diagonally dominant. To test for diagonal dominance, we use the following condition.

$$|a_{ii}| > \sum_{j=1, j\neq i}^{n} |a_{ij}|$$

Let us place each set of equations into matrix form and testing if the two conditions apply

Set 1:

$$\begin{bmatrix} 9 & 3 & 1 \\ -6 & 0 & 8 \\ 2 & 5 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \\ 6 \end{bmatrix}$$

Because a_{22} is a zero, this system of equations cannot be solved using the naive G-S algorithm

Set 2:

$$\begin{bmatrix} 1 & 1 & 6 \\ 1 & 5 & -1 \\ 4 & 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \\ 4 \end{bmatrix}$$

This system of equations does pass the first condition. Checking for diagonal dominance

$$\begin{bmatrix} 1 \\ 5 \\ |-2| \end{bmatrix} >? \begin{bmatrix} 1+6 \\ 1+|1| \\ 4+2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} >? \begin{bmatrix} 7 \\ 2 \\ 6 \end{bmatrix}$$

This matrix is not diagonally dominant as rows 1 and 3 violate the conditions for diagonal dominance. Therefore, G-S convergence is not guaranteed.

Set 3:

$$\begin{bmatrix} -3 & 4 & 5 \\ -2 & 2 & -4 \\ 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 1 \end{bmatrix}$$

This matrix does pass the first condition. Let us check for diagonal dominance.

$$\begin{bmatrix} |-3| \\ 2 \\ |-1| \end{bmatrix} >? \begin{bmatrix} 4+5 \\ |-2|+|-4| \\ 2 \end{bmatrix}$$
$$\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} >? \begin{bmatrix} 9 \\ 6 \\ 1 \end{bmatrix}$$

This matrix is not diagonally dominant as all rows violate the conditions for diagonal dominance.

<u>Part B:</u>

Set 1:

Pivoting is performed to ensure that there is a non-zero value in each diagonal element and that the diagonal element of each row is as large as possible. For set 1, this was accomplished by swapping row 3 with row 2.

$$\begin{bmatrix} 9 & 3 & 1 \\ 2 & 5 & -1 \\ -6 & 0 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 8 \end{bmatrix}$$

Testing the dominance condition yields

$$\begin{bmatrix} 9 \\ 5 \\ 8 \end{bmatrix} >? \begin{bmatrix} 3+1 \\ 2+|-1| \\ |-6| \end{bmatrix}$$
$$\begin{bmatrix} 9 \\ 5 \\ 8 \end{bmatrix} >? \begin{bmatrix} 4 \\ 3 \\ 6 \end{bmatrix}$$

This matrix is diagonally dominant. The naive G-S algorithm has guaranteed convergence.

Set 2:

Pivot row 3 with row 1

$$\begin{bmatrix} 4 & 2 & -2 \\ 2 & 5 & -1 \\ 1 & 1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 8 \end{bmatrix}$$

Testing the dominance condition yields

$$\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} >? \begin{bmatrix} 2+|-2| \\ 1+|-1| \\ 1+1 \end{bmatrix}$$
$$\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} >? \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}$$

The diagonal of all rows except the first is larger than the absolute sum of off-diagonal terms. However, the first row does show that the diagonal is equal to the absolute sum of off-diagonal terms. This does not strictly meet the conditions for diagonal dominance. However, the G-S algorithm may still converge in this case, as this row is so close to fulfilling the dominance criteria and the remaining rows also do. One would need to perform the naive G-S algorithm to be sure.

Set 3:

Pivoting is accomplished by swapping rows 2 and 3.

$$\begin{bmatrix} -3 & 4 & 5 \\ 0 & 2 & -1 \\ -2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ -3 \end{bmatrix}$$

Testing the dominance condition yields

$$\begin{bmatrix} |-3| \\ 2 \\ 4 \end{bmatrix} >? \begin{bmatrix} 4+5 \\ |-1| \\ |-2|+4 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} >? \begin{bmatrix} 9 \\ 1 \\ 6 \end{bmatrix}$$

No matter how pivoting is accomplished, this system of equations will not produce convergent behaviour using the naive G-S algorithm.

Question

Use least-squares regression to fit a straight line to the following data:

x	6	7	11	15	17	21	23	29	29	37	39
y	29	21	29	14	21	15	7	7	13	0	3

a) Calculate the slope and intercept of the straight line (linear) fit using linear regression.
b) Compute the coefficient of determination (r^2).
c) Plot the data and the regression line by hand (approximate is fine). If someone made an additional measurement of x = 10, y = 10, would you suspect, based on a visual assessment that the measurement may be an outlier?

Solution

Part A)

The equation of a line is be defined as

$$y = a_0 + a_1 x$$

Using linear regression, coefficients are given by

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \bar{y} - a_i \bar{x}$$

Computing the required intermediate values yields

$$n = 11$$

$$\bar{x} = 21.2727$$

$$\bar{y} = 14.4545$$

$$\sum x_i = 234$$

$$\sum y_i = 159$$

$$\sum x_i y_i = 2380$$

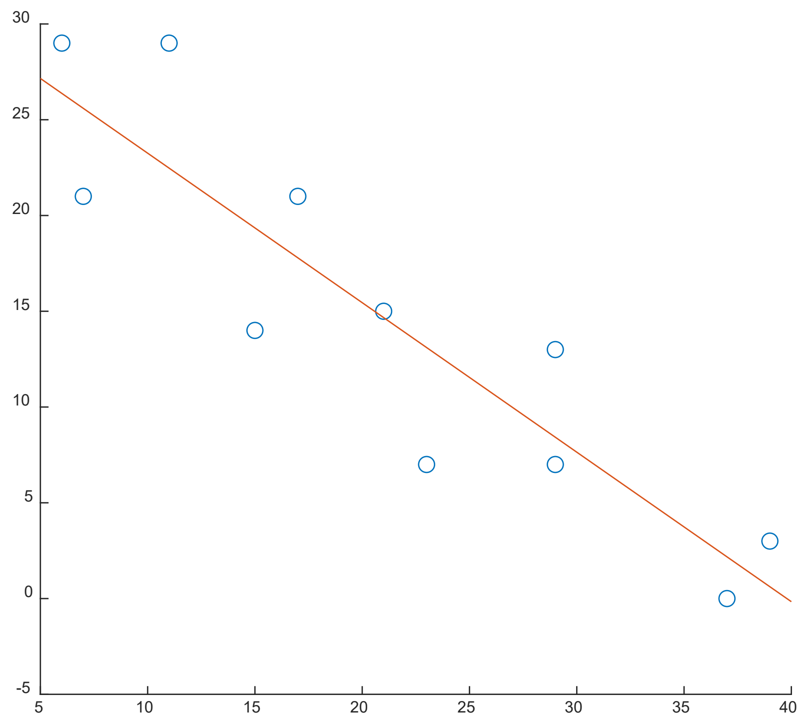
$$\sum x_i^2 = 6262$$

Computing the coefficients yields

$$a_1 = -0.7805$$

$$a_0 = 31.0589$$

This linear regression is plotted below.



Part B)

Compute the coefficient of determination r^2 using the following equation

$$r^2 = \frac{S_t - S_r}{S_t}$$

Where

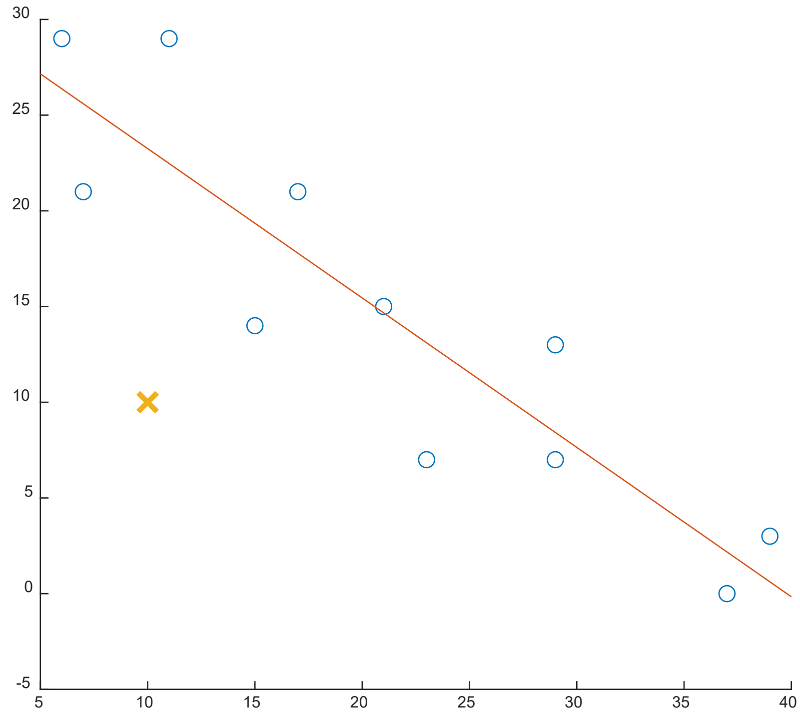
$$S_t = \sum (y_i - \bar{y})^2 = 962.7273$$
$$S_r = \sum (y_i - a_0 - a_1 x_i)^2 = 180.3358$$

This yields a coefficient of determination of r^2=0.8126

Part C)

Inserting x=10 into our linear fit produces a value of 23.2534, significantly larger than y=10. Furthermore, if we examine points around x=10, none have values close to y=10. This tells us the additional point would likely be an outlier.

Plotted against the existing data and the linear fit (red x), we can see that (10,10) is much lower than the rest of the data.



30
25
20
15
10
5
0
-5
5
10
15
20
25
30
35
40

Question

Evaluate the integral: ?_"0" ^"1" ?e^(x^2) dx using the trapezoid rule (n=4) and Simpson's 1/3 rule (n=4).

a) Trapezoid rule (with n=4)

Part I - Trapezoid $\int_0^1 e^{x^2}\,dx$

$$I = \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right]$$

Trapezoid rule with n=4

	upper	lower
Domain	1	0
h=(b-a)/n	0.25	
Trapezoid	**1.490679**	**ANSWER**

$$I = \underbrace{(b-a)}_{\text{width}} \underbrace{\frac{f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{Average height}}$$

n=	4

i	x	f(x)
0	0.00	1.00
1	0.25	1.06
2	0.50	1.28
3	0.75	1.76
4	1.00	2.72

a) Simpson's

n=	4

i	x	f(x)
0	0.00	1.00
1	0.25	1.06
2	0.50	1.28
3	0.75	1.76
4	1.00	2.72

Part II - Simpson's 1/3 rule

I = **1.463711**

$$I \cong \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4\sum_{i=1,3,5}^{n-1} f(x_i) + 2\sum_{j=2,4,6}^{n-2} f(x_j) + f(x_n)}{3n}}_{\text{Average height}}$$

b) Compare the results and discuss any differences in the estimates.

- The trapezoid rule has low accuracy in general (compared to other methods) and using only 4 intervals will make the error quite large. Recommended to used increased intervals and check for convergence of the solution.

Simpson's 1/3 rule is much more accurate (error is on the order of $1/n^4$)